Angiotensin-converting enzyme (ACE) inhibitors prevent the conversion of angiotensin I into angiotensin II. Angiotensin II is a powerful vasoconstrictor that increases aldosterone levels and stimulates the sympathetic nervous system. As angiotensin II levels decrease, the level of vasoconstriction lowers and aldosterone levels decrease, resulting in reduced sodium and water retention. ACE inhibitors also prevent the degradation of bradykinin, an endogenous vasodilator.

The most common side effect of ACE inhibitors is allergic rhinitis and cough, believed to be due to the increased levels of bradykinin it produces. Less common, but more serious reactions include angioedema and decreased glomerular filtration rate. Because ACE inhibitors reduce aldosterone levels, hyperkalemia may occur. The risk of hyperkalemia is greatest in patients with congestive heart failure and renal insufficiency.

Drug	Dose (mg)	Duration (hours)
Captopril	100	6-10
Enalapril	20	18-30
Lisinopril	10	18-30
Ramipril	20	24-60

The most pronounced adverse effect of ACE inhibitors during anesthesia is prolonged hypotension. Because of this, it is recommended that patients on ACE inhibitors discontinue therapy at least 12-24 hours prior to surgery. If hypertension becomes an issue, then they may need to be treated with shorter-acting agents until discharge. If a patient on ACE inhibitors exhibits hypotension that is refractory to intravenous fluid boluses, ephedrine, and/or phenylephrine, then vasopressin or the long-acting form of vasopressin, terlipressin, should be used.

Stoelting RK & Hillier SC. Pharmacology and Physiology in Anesthetic Practice. 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006: 345-347.

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